

1. Why python is dynamically typed language.

Ans:

Python is called a dynamically typed language because you do not need to declare the data type of a variable when creating it. Python determines the type automatically at runtime, and the same variable can later refer to a value of a different type.

Example

```
x = 10
```

```
print(type(x))
```

Here, Python automatically identifies x as an integer (int).

Later:

```
x = "Hello"
```

```
print(type(x))
```

Now x refers to a string (str).

So, the type can change during program execution.

Simple answer

Python is dynamically typed because variable types are determined at runtime, and we do not need to explicitly declare the data type of a variable. A variable can also refer to values of different data types during execution.

Remember:

Dynamic typing = Type is decided at runtime.